CS1231S: Discrete Structures

Tutorial #11: Graph II and Tree

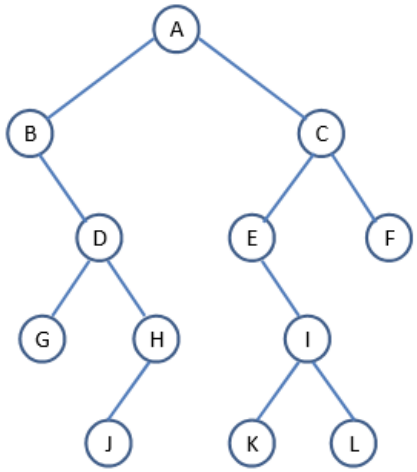
(Week 13: 9 – 13 November 2026)

I. Discussion Questions

D1. For any simple connected graph with n $(n > 0)$ vertices, what is the minimum and maximum number of edges the graph may have?

D2. (AY2016/17 Semester 1 Exam Question)
How many simple graphs on 3 vertices are there?
In general, how many simple graphs on n $(n > 1)$ vertices are there?

D3. Given the binary tree on the right, write the pre-order, in-order, and post-order traversals of its vertices.



Note: Monday's groups are affected by the public holiday on 9 November 2026. A mass online make-up tutorial on Zoom will be conducted on Thursday 12-1:40pm (the usual CS1231S lecture time slot) that week for the Monday's groups.

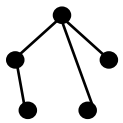
II. Tutorial Questions

You can freely use the following lemma (without proof). The proof is left as an optional exercise.

Lemma 10.5.5. Let G be a simple, undirected graph. Then if there are two distinct paths from a vertex v to a different vertex w, then G contains a cycle (and hence G is cyclic).

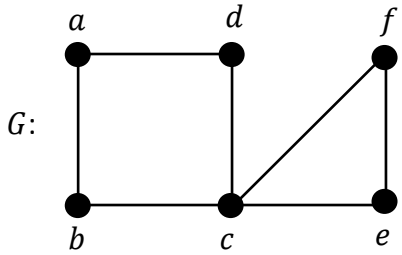
1. (a) For the following graph G, draw its complement graph $\bar{G}$.

G:



(b) Consider simple graphs on n vertices. Draw all self-complementary graphs with n vertices (for $n = 3, 4, 5, 6$), or justify why there are none.

2. Consider the graph G given below. How many spanning trees of G are there?



3. (a) Draw all non-isomorphic trees with n vertices, $n = 1,2,3,4$.

   (b) For each non-isomorphic tree in part (a), how many isomorphic copies are there?

4. (a) Let $G = (V, E)$ be a simple, undirected graph. Prove that if G is connected, then $|E| \geq |V| - 1$.

   (b) Is the converse true?

5. (a) Let $G = (V, E)$ be a simple, undirected graph. Prove that if G is acyclic, then $|E| \leq |V| - 1$.

   (b) Is the converse true?

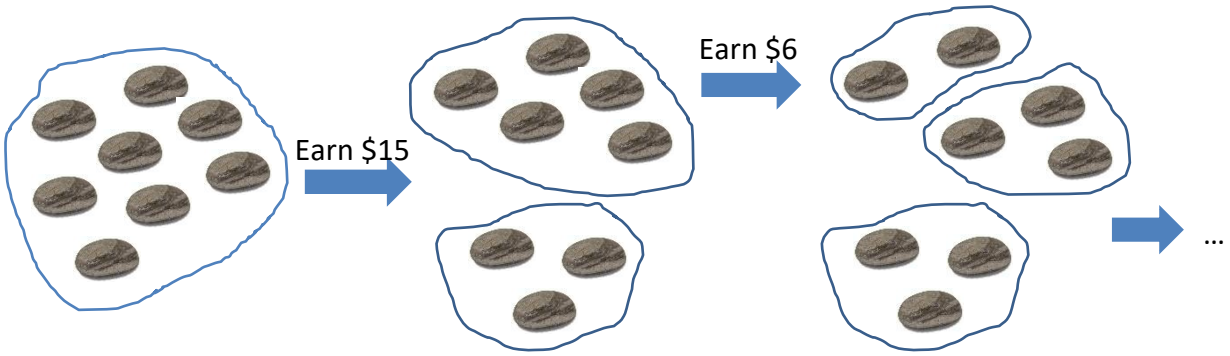
6. Let $G = (V, E)$ be a simple, undirected graph. Prove that G is a tree if and only if there is exactly one path between every pair of vertices.

7. (AY2017/18 Semester 1 Exam Question)
   Suppose you are given a pile of stones. At each step, you can separate a pile of k stones into two piles of k_1 and k_2 stones. (Obviously, $k_1 + k_2 = k$.) On doing this, you earn $\$(k_1 \times k_2)$.

   What is the maximum amount of money you can earn at the end if you start with a pile of n stones? Explain your answer.

   The diagram below illustrates the (incomplete) process of separating a pile of 8 stones.

   (This problem may be solved without using graph theory, but here we attempt to model it as a graph problem.)



8. (a) A binary tree T_1 has 9 vertices. The in-order and pre-order traversals of T_1 are given below. Draw the tree T_1 and give its post-order traversal.

In-order: E A C K F H D B G
Pre-order: F A E K C D H G B

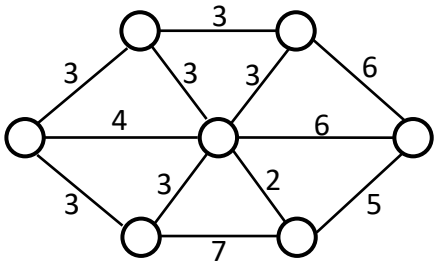
(b) A binary tree T_2 has 9 vertices. The in-order and post-order traversals of T_2 are given below. Draw the tree T_2 and give its pre-order traversal.

In-order: D B F E A G C H K
Post-order: D F E B G K H C A

9. (a) Draw all possible binary trees with 3 vertices X, Y and Z with in-order traversal: $X\ Y\ Z$.

(b) Draw all possible binary trees with 4 vertices A, B, C and D with in-order traversal: $A\ B\ C\ D$.

10. (AY2023/24 Semester 2 Exam Question)
An undirected weighted graph is shown below.



(a) What is the weight of its MST?

(b) How many non-isomorphic MSTs are there? Draw them out.

All The Best For Your Final Exams!

